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Embedded IoT Platform for Intelligent Living Space

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ABSTRACT

Ensuring safety, automation, and energy efficiency in contemporary living spaces has become a pressing concern owing to the increasing safety risks, energy requirements, and need for real-time monitoring. This paper proposes the design and development of an Embedded IoT Platform for Intelligent Living Space using distributed ESP32 nodes. The proposed work aims to provide multi-sensor monitoring, real-time local processing using cloud connectivity via RemoteXY for remote monitoring and notification. The system continuously monitors environmental factors like temperature, gas levels, motion sensing, soil moisture, and water tank levels. Based on predefined thresholds, automated functions like exhaust fan control, window operation, irrigation control, and smart door locking are carried out. Unlike traditional cloud-based systems, the proposed hybrid model ensures continuous functionality through local decision-making. The proposed system is modular, scalable, cost-effective, and appropriate for urban and semi-automated residential settings.

Keywords:- IoT, Smart Home, ESP32, RemoteXY, Home Automation, Sensor Networks

INTRODUCTION

Increasing Need for Smart Homes:

The development of IoT technology has greatly changed the traditional residential setting into a smart living space. Smart homes use sensors, microcontrollers, and communication systems to automatically monitor and control various parameters in a house.

Shortcomings of Traditional Homes:

Most traditional homes do not have real-time monitoring systems to detect dangers like gas leaks, fire breakouts, unauthorized entry, and improper water management. These systems are mostly dependent on human observation, which can be dangerous during emergencies.

Need For an Integrated Automated Homes:

There is a great need for a centralized and distributed system that continuously monitors environmental parameters and performs automated control tasks without the need for constant human intervention.

Proposed Solution:

The proposed solution presents a distributed ESP32- based IoT platform that supports multi-sensor to the monitoring, local decision-making, and cloud-based remote access for ensuring safety, comfort, and energy efficiency.

LITERATURE SURVEY

The history of home automation can be traced by examining the development of the communication technologies that support it. A literature review shows that while conventional home automation systems used Global System for Mobile Communications (GSM) or wired communication, current home automation systems are characterized by their flexibility, scalability, and ability to work together using platforms such as Wi-Fi, ZigBee, and Message Queuing Telemetry Transport (MQTT).

Communication Protocol Landscapes

The prerequisite for the security and confidentiality of smart home systems is wireless communication. Various network platforms have been compared by researchers to establish their applicability for particular automation applications. Wi-Fi (IEEE 802.11 b/g/n) is the most widely used platform because of its high data transfer rate and the widespread use of home routers, which serve as the natural gateway for IoT devices. However, ZigBee and Bluetooth Low Energy (BLE) are preferred for low- power sensor applications that require mesh network support.

The Transition to AIoT and Predictive Modeling

One of the most important trends that have been identified in the latest peer-reviewed articles (2024-2025) is the convergence of Artificial Intelligence (AI) and IoT, giving birth to the Artificial Intelligence of Things (AIoT). The older systems were mostly reactive, carrying out commands based on the crossing of static thresholds. However, the latest smart home systems use machine learning algorithms to predict and learn user behavior.

METHODOLOGY

A good smart home system has a logical framework that ensures a smooth flow of information from the physical world interface. The most advanced smart home systems have a modular framework that is categorized into four layers: perception, network, fog/middleware, and application.

The Perception Layer:

Data Acquisition and Execution

The perception layer is the base of the framework and is made up of physical components that can communicate with the environment. This layer is further categorized into sensors that are responsible for monitoring the environment by converting physical inputs

into information signals and actuators that execute commands to change the environment.

The sensor network may comprise:

Thermal Comfort Sensors: These sensors monitor temperature and humidity to provide a comfortable indoor environment.

Safety and Hazard Sensors: These sensors detect smoke, gas leaks (LPG and methane), and flames to avoid home accidents.

Security Sensors: These sensors comprise PIR motion sensors and door magnetic sensors to detect security breaches.

Visual and Environmental Sensors: These sensors include LDR sensors for home lighting and MQ-135 sensors for air quality measurement.

The actuators consist of relay modules used for switching on/off high-voltage devices, solenoid valves used for gas control, and electronic locks used for controlling doors.

The Network and Middleware Layers: Connectivity and Processing

The network layer is the connectivity framework necessary for communication between the perception layer and the central control units. The establishment of a strong and secure connection is vital for successful automation. In most contemporary systems, there is an intervening "Fog Computing" or "Edge Intelligence" component. This component, usually achieved through an ESP32, receives information from smaller network nodes and provides communication in case the main connection to the cloud is interrupted.

The network layer generally adheres to the OSI model, ensuring that information is retrieved, decoded into bits, and sent over the internet in packets. This layered system ensures low latency and that essential safety operations, such as turning on a

local buzzer in case of a gas leak, are accomplished without waiting for a round-trip response from the cloud.

The Application Layer: User Interaction and Monitoring

The application layer is the interface between the system and the human resident. It offers a centralized platform to control all existing home appliances remotely. The interaction options are:

Mobile Applications: Custom-built Android or iOS apps (e.g., RemoteXY) for real-time control and notification.

I. Technical Design and Microcontroller Specifications

The choice of a microcontroller unit (MCU) dictates the computational power, connectivity options, and scalability of the smart home system. Research papers frequently compare three main platforms ESP32

ESP32 - The Core of Wireless Sensing Nodes:

The ESP32 is an open-source IoT platform that combines the ESP32 Wi-Fi SoC with 4MB of flash memory. It is highly valued in smart home research for its integrated Wi-Fi connectivity and low cost, which significantly simplifies the peripheral circuit requirements compared to traditional Arduino-based systems.

ESP32 - Improved Connectivity and Edge Computing:

The ESP32 is the successor to the ESP8266 and is increasingly being used in 2024-2025 projects due to its dual-core processor and the ability to connect using both Wi-Fi and Bluetooth. With a clock speed of up to 240 MHz and a performance level of up to 600 DMIPS, the ESP32 is able to handle edge computing tasks, where initial processing of the data is done before uploading the refined packets to the cloud.

ESP32 – CAM Module:

ESP32 is a low cost high speed WIFI + Bluetooth Microcontroller with a build camera module based on the ESP32 SOC and its typically paired with OV3640 camera sensor used for Face recognition Live streaming and Security Purpose.

It is a dual core 32-bit LX6 with clock speed upto 240MHz and PSRAM 4MB (Important for high resolution). Having 9 Usable GPIO Pins and for storing data an Micro SD Card Slot. With Operating Voltage 5V.

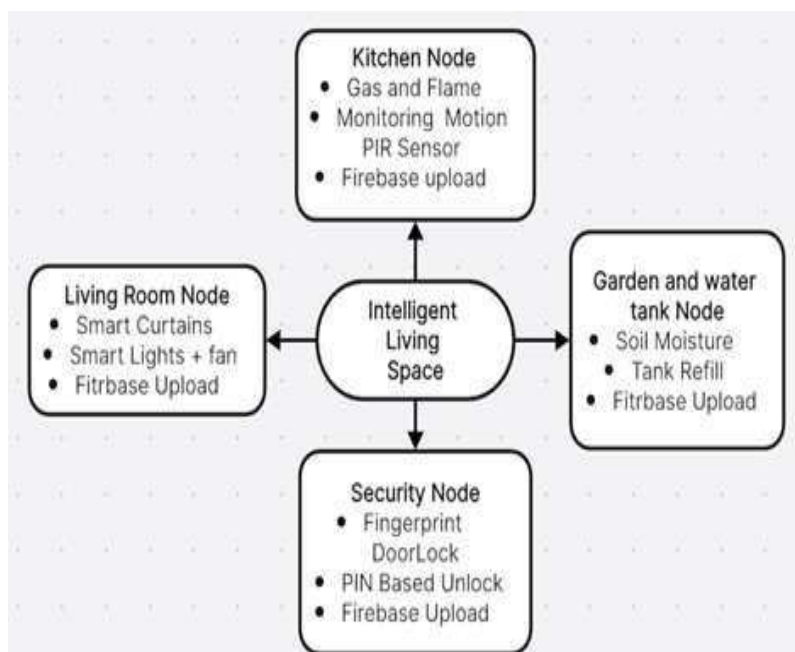


Fig.1:-Proposed Embedded IoT Architecture for Intelligent Living Space

II. Sensor Modularity and Signal Acquisition Characteristics

The reliability of a smart home platform is directly proportional to the accuracy and response time of its sensing components. Each sensor module employs unique physical principles for sensing environmental changes.

DHT11: Environmental Thermal Sensing

The DHT11 sensor is a popular choice for sensing air temperature and humidity. The DHT11 sensor has a calibrated digital signal output, which ensures high reliability and stability. It employs a digital signal acquisition technique that includes a resistive-type humidity sensing component and an NTC temperature sensing component.

The communication process for the DHT11 is done through a single-wire serial interface. MCU sends a start signal, which triggers the DHT11 to send a 40-bit data packet (8-bit integral RH + 8-bit decimal RH + 8-bit integral T + 8-bit decimal T + 8-bit checksum). The whole communication process takes around 4ms. MQ Gas Sensor Series: Hazard and Smoke Detection.

The MQ series sensors are meant for detecting different types of combustible and toxic gases. The functioning of the sensors is based on the principle of detecting variations in electrical resistance due to the interaction of gas molecules with a heated Tin Dioxide (SnO₂) layer.

MQ-2: Highly sensitive to smoke, LPG, methane, and hydrogen. It is widely used for domestic safety and industrial applications.

MQ-6: Specifically designed for LPG, isobutane, and propane. It is regarded as better than MQ-2, specifically for gas leak detection, due to its higher sensitivity to LPG concentrations.

Motion and Light Detection

PIR (Passive Infrared): Detects human motion by measuring the infrared radiation emitted by objects within its range. It is critical for presence-controlled lighting and security systems.

LDR (Light Dependent Resistor): The resistance level drops as the light intensity increases. It is used in curtain and lighting control systems to maximize visual comfort and energy efficiency.

Soil Sensor :

A capacitive soil moisture sensor is used to measure the moisture level in the soil based on the principle of capacitance. The sensor detects changes in capacitance caused by variations in soil water content, as water has a higher dielectric constant than dry soil. It provides an analog output voltage corresponding to the moisture level, which can be read by a microcontroller.

Ultrasonic Sensor

An ultrasonic sensor is used to measure the water level without direct contact with the liquid. It works by transmitting ultrasonic sound waves toward the water surface and measuring the time taken for the echo to return. Based on this time, the distance between the sensor and the water surface is calculated. This distance is then used to determine the water level in the tank.

Monitoring Platforms and Cloud Infrastructure

The success of an IoT sensing platform

depends on its capacity to consolidate data for analysis. Cloud infrastructure has the capacity to store, process, and display data for meaningful analysis of raw sensor data.

RemoteXY: Open-Source Energy and Environment Monitoring

RemoteXY is a mobile application used to control and monitor microcontroller-based systems wirelessly through a smartphone. It allows users to create a custom interface with buttons and displays to interact with devices like Arduino and ESP8266.

RemoteXY communicates with the microcontroller using Wi-Fi or Bluetooth, enabling remote control and real-time monitoring of sensors and devices. It is widely used in home automation, irrigation systems, and IoT projects due to its simplicity and ease of use.

IOT Android Integration :

For rapid development and user-friendly applications, the Remotexyz platform provides an easy-to-use mobile interface. Users can use the drag-and-drop widget feature to design control interfaces for relays and display interfaces for sensor data. Blynk enables secure notification and alarm messaging for dangerous situations such as burglary and LPG leakage.

Secure Remote Access and Gateway Technology :

A major issue in cloud-based systems is latency and connectivity issues. Researchers have used Cloudflare to guarantee that remote access is both responsive and secure from unauthorized breaches. To enhance security, some systems use local storage on a dedicated server to guarantee that all sensitive information remains on-site and never leaves the premises.

IMPLEMENTATION AND PERFORMANCE EVALUATION

Scientific verification of a smart home platform demands thorough testing on various performance parameters. A Network Latency and Responsiveness Latency tests verify the system's efficiency in emergency situations.

Energy Efficiency Benchmarks

The integration of intelligent sensors allows for substantial energy savings by optimizing device usage based on real-time demand and environmental condition. These are made possible by functionalities such as "smart user behavior lighting," which turns off lights if there is no motion for a set period of time, and automatic HVAC control based on the presence or absence of occupants.

Sensor Accuracy and Reliability Testing

The results of experimental testing on developed prototypes indicate a high level of reliability in sensor data. The temperature sensor (DHT11) has an average deviation of only 0.48% during a testing phase, while light and gas sensors have deviations of 0.58% and 0.42%, respectively. This accuracy is crucial to ensure that the system offers a reliable interface to the user.

User Satisfaction and Usability

User satisfaction surveys carried out for a period of one month on prototype systems have shown a high level of user satisfaction (8.7/10), particularly in terms of ease of use and the user-friendly nature of mobile dashboards. The ISO 25010 standard has been employed to measure the effectiveness of the system, and one study has indicated an outstanding performance rating of 4.70 out of 5 in terms of categories such as compatibility, reliability, and secure.

RESEARCH TRENDS

The smart home research environment in

2025 is characterized by four major pillars: Edge AI, Blockchain security, ethical frameworks for AI, and sustainable energy management.

The Emergence of Edge AI

To overcome the challenges of cloud computing, there is a growing need to move machine learning closer to the data source, a process known as "Edge Intelligence" or "Edge AI". This enables real-time anomaly detection and predictive maintenance on IoT devices such as the ESP32 using state-of-the-art machine learning and deep learning algorithms without the latency of cloud offloading.

Ethical Frameworks for AIoT

With the increasing integration of AI in the home, ethical issues related to data privacy, transparency, and human autonomy have become prominent. Current research uses the User Requirements Notation (URN) framework to study the connection between AI technology and ethics.

The following principles are being incorporated into contemporary designs:

Privacy by Design: Data protection as a fundamental capability from inception.

Explainability: Giving users meaningful explanations for an automated decision.

Accountability: Assigning responsibility for automated decisions in smart home systems.

Blockchain for Decentralized Security

To counter the risk of hacking attacks, blockchain technology is being considered as a solution for securing smart home networks. Blockchain technology will help create a decentralized system for authenticating devices, allowing only authorized devices to communicate within the smart home environment and preventing any unauthorized manipulation of communication protocols.

Sustainable Energy Management and PV Integration

Current research in 2025 involves the integration of IoT technology with renewable energy resources. This includes tracking the parameters of a photovoltaic (PV) cell (voltage, current, and power) and automatically allocating high-power tasks when the solar energy output is at its peak. This "Energy-Aware Control" is critical in making smart homes environmentally friendly.

CONCLUSION

The thorough examination of the IoT-based smart home automation system has made it clear that the technology is at control to proactive and intelligent automation systems through AIoT and edge computing. The technical assessment has shown that the platforms developed using NodeMCU-ESP8266 and ESP32 have reached a high level of reliability, with the accuracy of temperature and gas sensors close to commercial standards.

However, as the systems become more autonomous, the research community must also address the challenges of interoperability among different vendors and the ethical implications of data collection. The future will likely include the integration of 6G-enabled IoT, green energy harvesting for battery-less sensors, and the development of more explainable AI models that improve user trust.

By following strict architectural guidelines and academic reporting, the development of these sensing and monitoring systems will continue to play a crucial role in the development of a safer, more efficient, and more sustainable living space for the global user.

The research roadmap from prototype to peer-reviewed publication not only demands technical creativity but also a commitment to standardized research

methodology and the search for solutions that meet the changing needs of the modern user.

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